

MINSEOK PARK

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RESEARCH INTERESTS

UAV Autonomy, Robust Path Planning, Reinforcement Learning

EDUCATION

B.S. Electronic Engineering | Hanyang University

- GPA 3.84/4.0

Seoul, Republic of Korea

Expected Feb 2027

Exchange Program Electrical Engineering | University of North Dakota

- GPA 4.0/4.0

Grand Forks, ND

Jan - May 2026

RESEARCH EXPERIENCE

Undergraduate Researcher (Advisor: Prof. Jueming Hu)

University of North Dakota

- Resilient UAV Navigation under Sensor Dropout**

Jan 2026 - Present

- Designed and implemented a simulation environment, model architecture, training, and evaluation framework for resilient UAV navigation under intermittent sensor dropout.
- Achieved up to 91-95% task completion rate across localization and range sensor failure conditions.
- Manuscript in preparation, targeting *IEEE IoT*.

- Robust Fault Diagnosis of Rotating Machinery via Entropy-Augmented CNN**

Jan - May 2026

- Designed and implemented a pipeline including data preprocessing, model architecture, training, and evaluation framework for entropy-based CNN fault diagnosis.
- Achieved over 95% classification accuracy across both time and frequency domains.
- Accepted at ASME IMECE 2026.

- End-to-End RL-Based UAV Waypoint Navigation**

Jan - May 2026

- Augmented the state space of an existing circular path-following controller with goal position, developing an end-to-end RL policy for single-waypoint hovering navigation in a Crazyflie PyBullet simulation environment.
- Progressed from single-waypoint hovering to sequential multi-waypoint navigation with randomized goal spawning.

Coursework Project

Hanyang University

- Multi-Anchor CNN with ToA Attention for UWB Indoor Localization**

Sep - Dec 2025

- Designed and implemented a Multi-Anchor CNN integrating ToA-based Softmax Attention to suppress NLoS interference for UWB-based indoor localization.
- Achieved RMSE of 1.439m, below the 1.5m commercialization target.

PUBLICATION

Robust Fault Diagnosis of Rotating Machinery Using Entropy-Augmented CNN in Time and Frequency Domains

August 2026

ASME International Mechanical Engineering Congress and Exposition (IMECE) 2026

Accepted

- First author, Responsible for data preprocessing, model implementation, experimental evaluation, and manuscript drafting.

AWARDS

Korea-U.S. Advanced Technology Youth Exchange Support Program

2026

Korea Institute for Advancement of Technology (KIAT)

- 4th Batch, Aerospace Track

College of Engineering Leadership Scholarship

2025

Hanyang University

SKILLS

Programming: Python (PyTorch)

AI & Machine Learning: CNN, Reinforcement Learning (RL), Imitation Learning (IL)

Robotics & Simulation: PyBullet, Crazyflie, OpenAI Gym

Languages: Korean (Native), English (Proficient)

MILITARY EXPERIENCE

Sergeant

Feb 2022 - Nov 2023

Republic of Korea Air Force

- Completed mandatory military service (Honorable Discharge).